RIDWAN AMURE

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EDUCATION

University of Arkansas at Little Rock — Little Rock, AR
PhD, Computer Science — *Expected May 2027*
MS, Computer Science — *May 2025*

EXPERIENCE

Meta — *Aug 2026 – Present*
Part-time Student Researcher, Ads Ranking (Early Stage Ranking)
- Improving ESR performance across all production metrics through target-aware attention, including hard inductive bias via ID matching and cross-attention.

Meta — *June 2026 – Aug 2026*
Software Engineering Intern, Ads Ranking (Early Stage Ranking)
- Applied product quantization and resolved model-lowering/inference compilation blockers, enabling GPU-based ESR serving at billions-of-candidates scale with improved latency.
- Developed a GRPO-based auxiliary loss for ESR (adapting LLM post-training optimization to ranking) and designed hard target-aware attention via ID matching, yielding gains in Normalized Entropy and recall.
- Introduced joint retrieval with in-batch softmax loss for ESR, improving training signal efficiency for large-scale candidate retrieval.

Meta — *May 2025 – Aug 2025*
Software Engineering Intern, Machine Learning
- Built an end-to-end ML data pipeline reducing training latency from 14 days to 1 day; improved ranking model quality through feature engineering, ablation studies, and model-based retrieval.
- Designed a transformer-based interaction architecture for retrieval and ranking, optimizing inference-time performance at scale.

University of Arkansas at Little Rock — *Aug 2023 – Present*
Graduate Researcher, COSMOS Research Center
- Built large-scale AI pipelines for cross-platform social media analysis, processing millions of posts across 100K+ user datasets.
- Developed knowledge graph and embedding-based models for narrative tracking; designed ML systems for information diffusion modeling.

IQuartic (NewWave Company) — *Jan 2023 – Aug 2023*
Senior Machine Learning Engineer
- Developed end-to-end ML systems for ICD-to-HCC risk adjustment coding, increasing financial gains by over 30% through improved coding accuracy.
- Built Bayesian inference models to identify missed or suboptimal medical coding across large-scale electronic health record datasets.
- Designed NLP and RAG pipelines for structured extraction of clinical text; managed full model lifecycle from training through production monitoring.

ParallelScore — *Sep 2021 – Apr 2023*
Data Science Engineer
- Developed transformer-based NLP models for medical coding (PHI detection, ICD extraction, visit classification), improving PHI detection from <50% to 95% in QA and 89% in production.
- Built scalable ML pipelines for preprocessing and structured insight extraction from large-scale unstructured clinical text.
- Designed and deployed real-time inference systems using AWS, Docker, and Kubernetes; conducted model evaluation and performance optimization.

Babban Gona — *Sep 2020 – Sep 2021*
Machine Learning Engineer
- Built and deployed computer vision models for agricultural diagnostics, reaching 200,000+ users in resource-constrained field environments.

- Developed low-latency inference systems for real-time plant condition detection; maintained and optimized production ML systems at scale.

SKILLS

Python, PyTorch, TensorFlow, Transformers, Ranking Systems, RAG, Agentic AI, Knowledge Graphs, AWS, Docker, Kubernetes